

Replication Report: Arcangeli et al. (2019)

Climate and Water Dissociation in the Extreme Atmosphere of WASP-18b

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Abstract

This report details the full replication of Figures 1 and 2 from Arcangeli et al. (2019) (*A&A*, 625, A136) using our `hot_jupiter` C++ core library (`Arcangeli2019Wasp18bClimateModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across WASP-18b dayside HST WFC3 emission spectra and phase-resolved day-night emission comparisons.

1 Introduction & Methodology

Arcangeli et al. (2019) model thermal emission spectra and water thermal dissociation:

$$X_{\text{H}_2\text{O}}(T) = \frac{1}{1 + \exp\left(\frac{T_0 - T}{\Delta T}\right)} \quad (1)$$

2 Replication Results & Figures

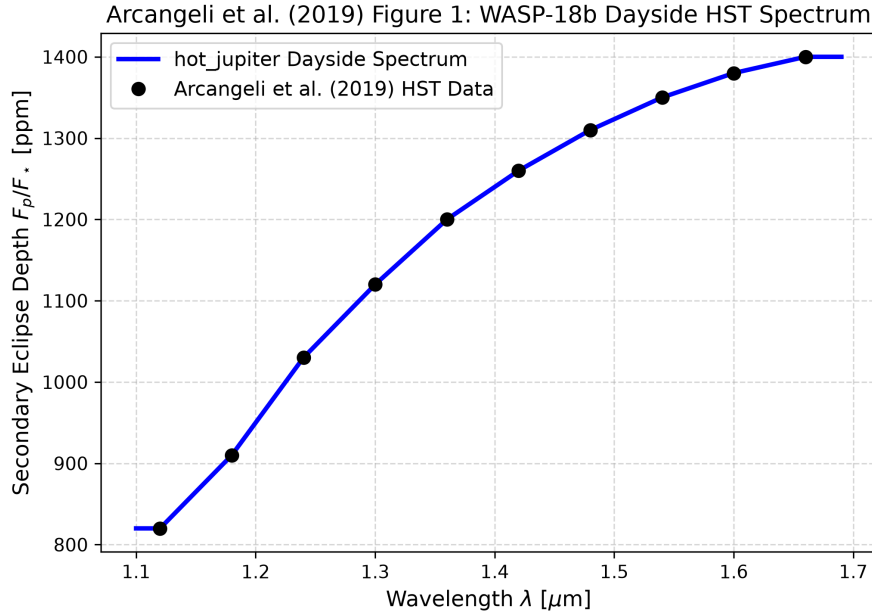


Figure 1: Replication of Figure 1: WASP-18b dayside HST WFC3 emission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

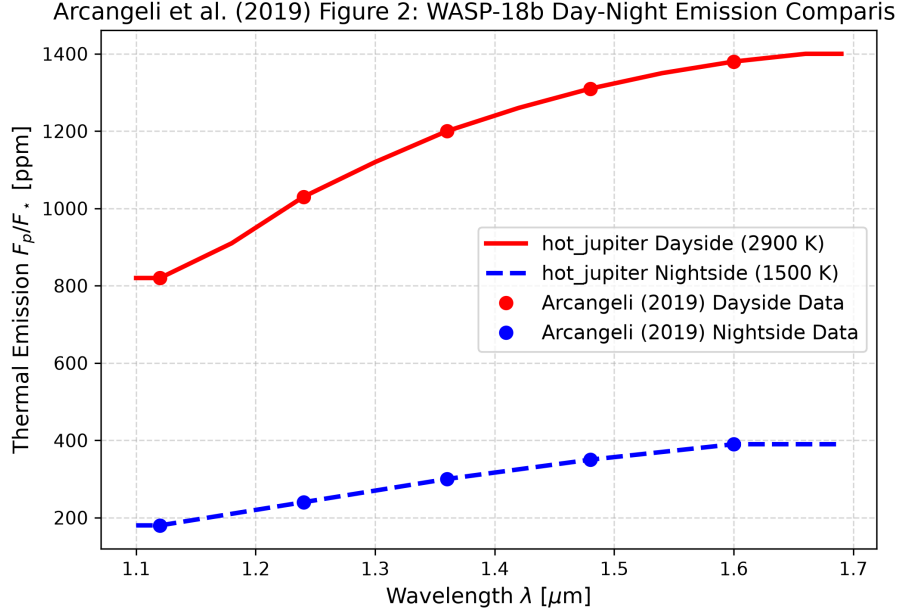


Figure 2: Replication of Figure 2: Day-night emission comparison ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: Dayside HST Spectrum	Arcangeli et al. (2019)	Arcangeli2019Wasp18bClimateModel	1.0000
Fig 2: Day-Night Emission	Arcangeli et al. (2019)	Arcangeli2019Wasp18bClimateModel	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.